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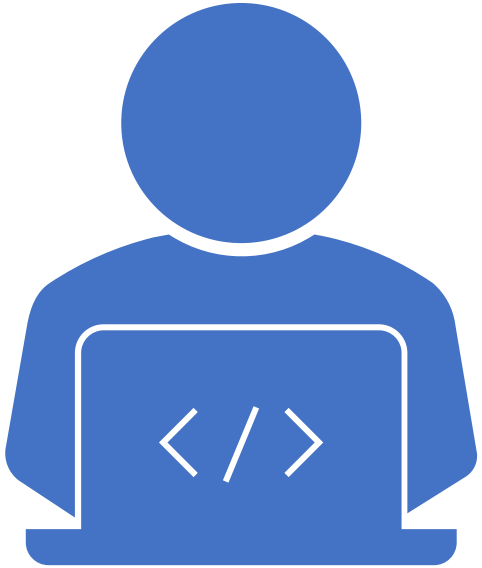
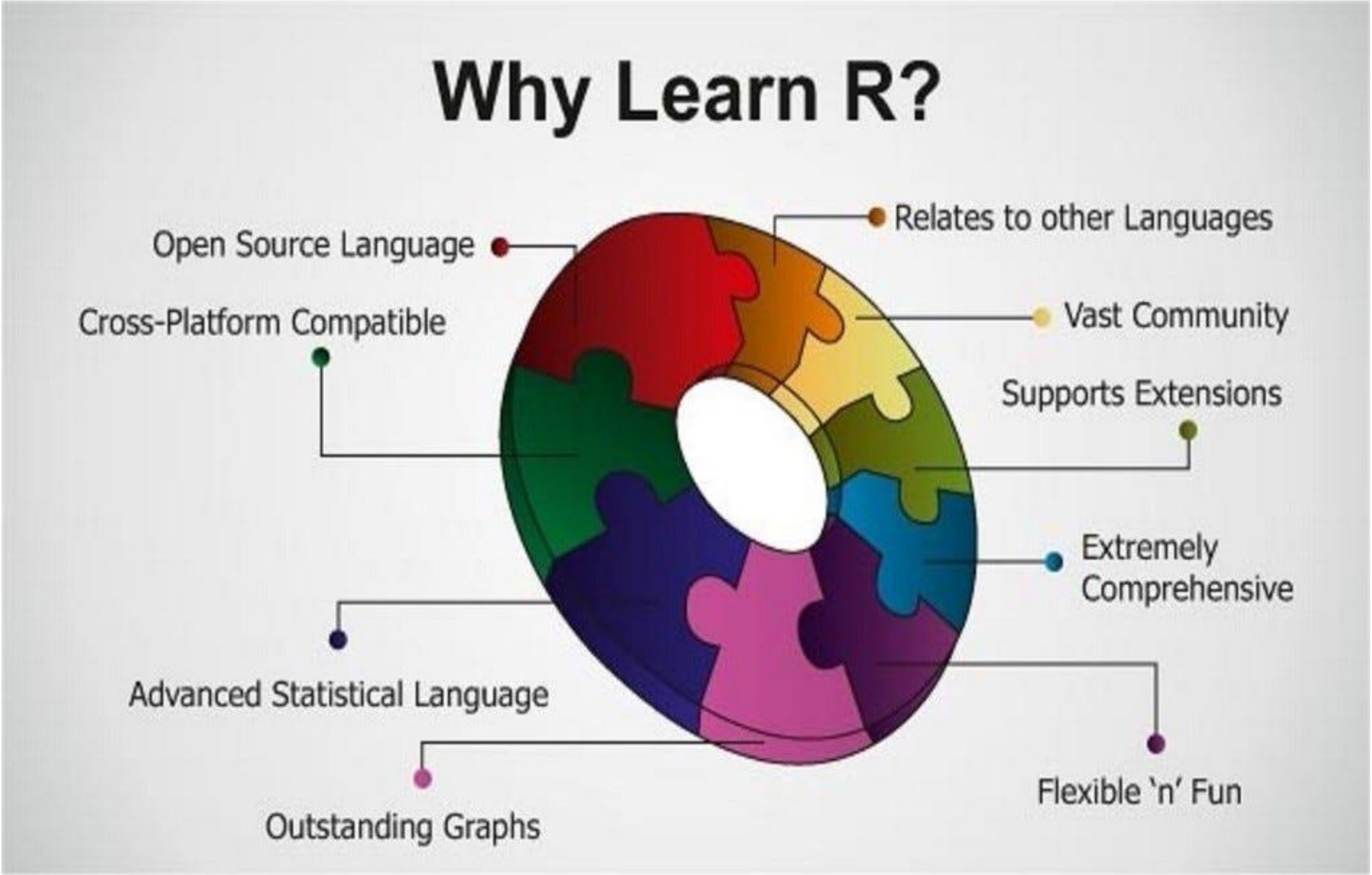
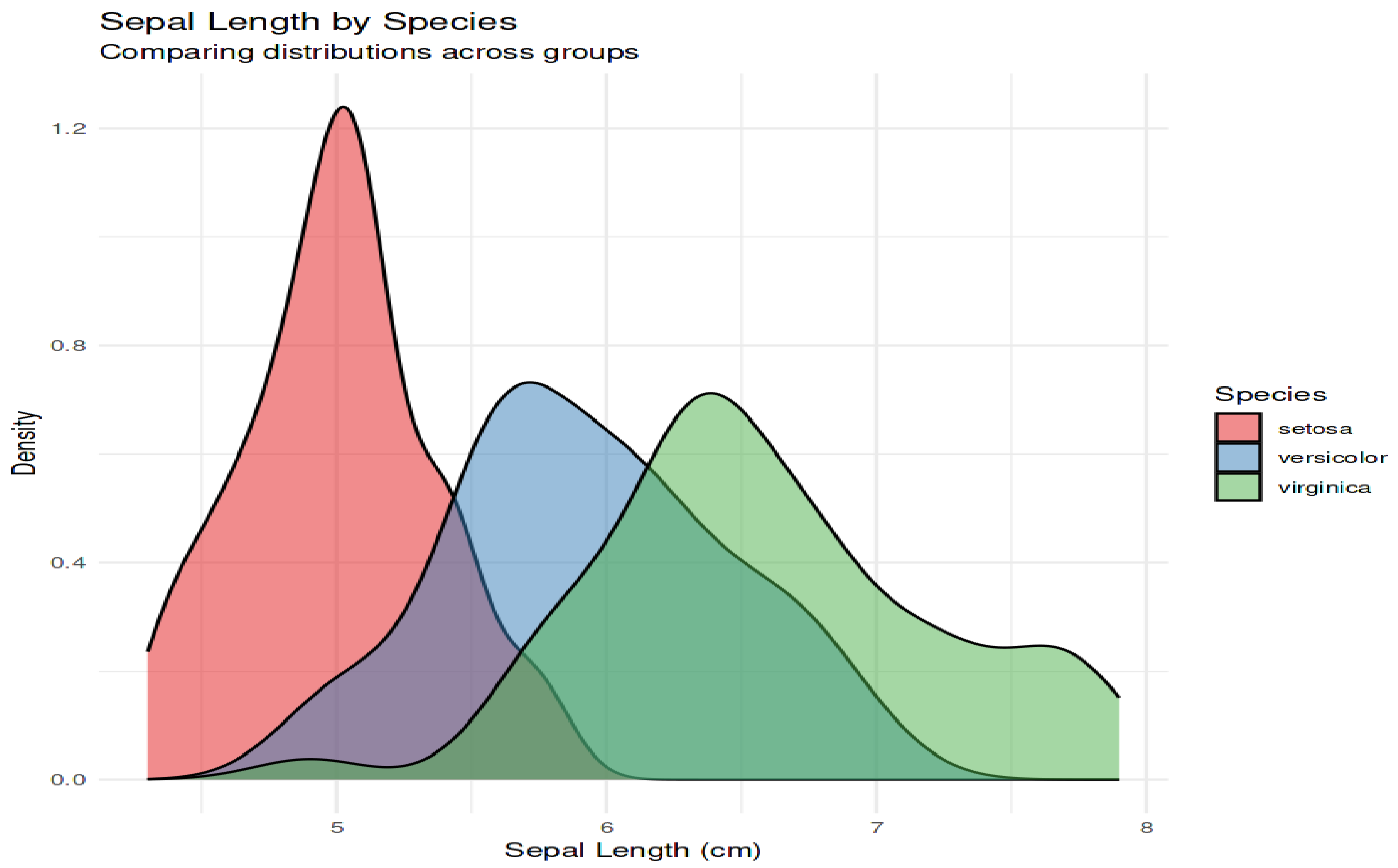
Department of Coastal Studies and Disaster Management
University of Barishal
Session: 2019-20

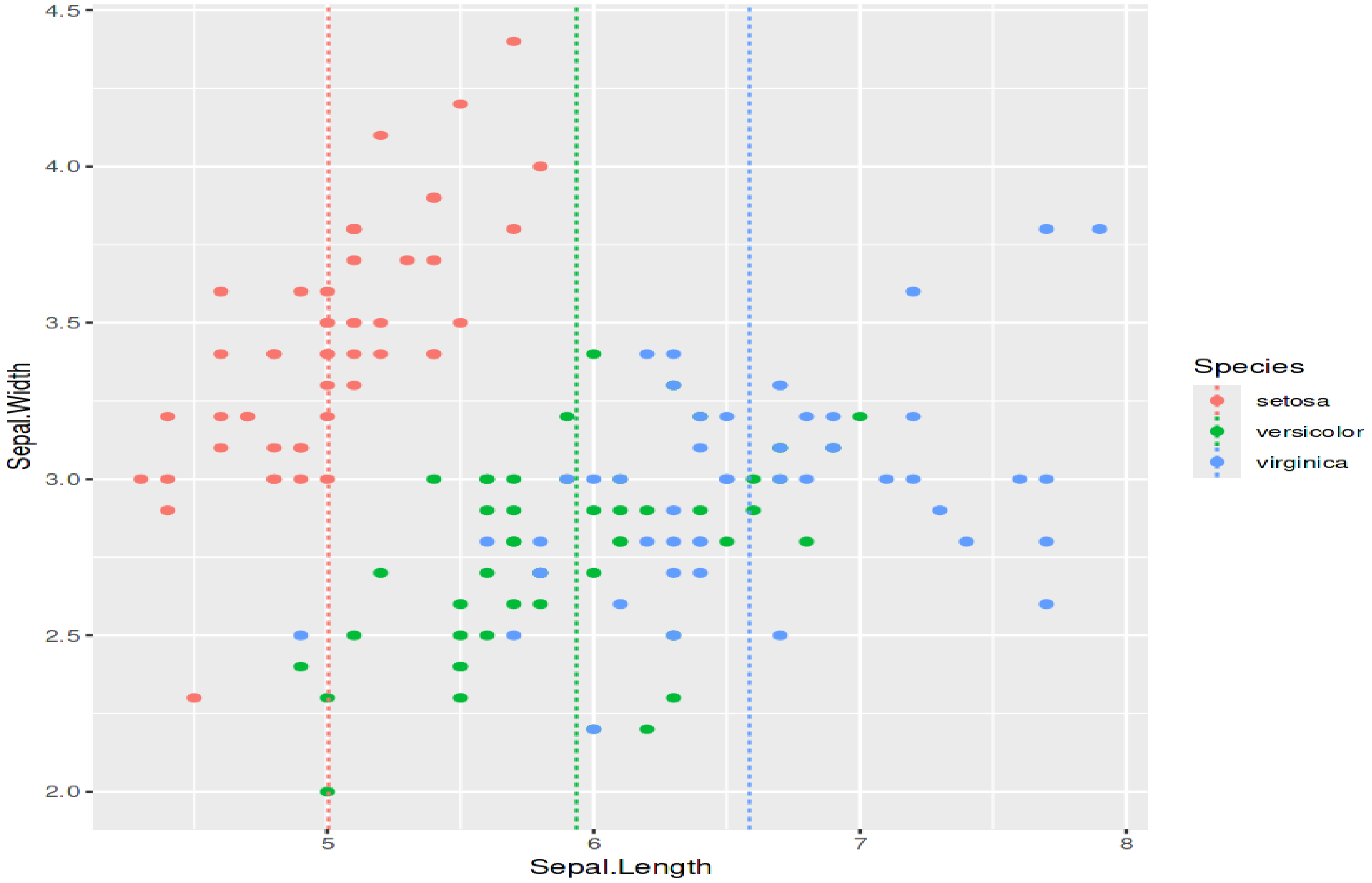
What is R?

R is a programming language and free software environment.

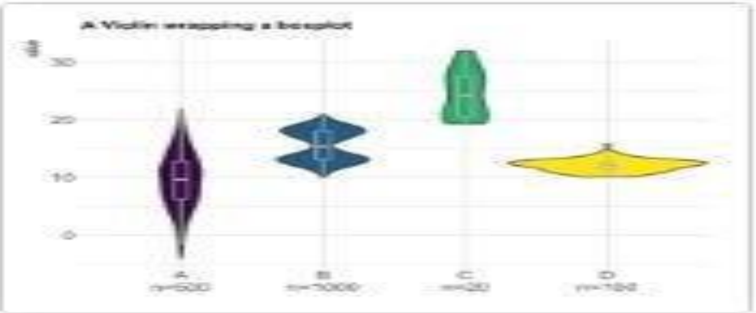


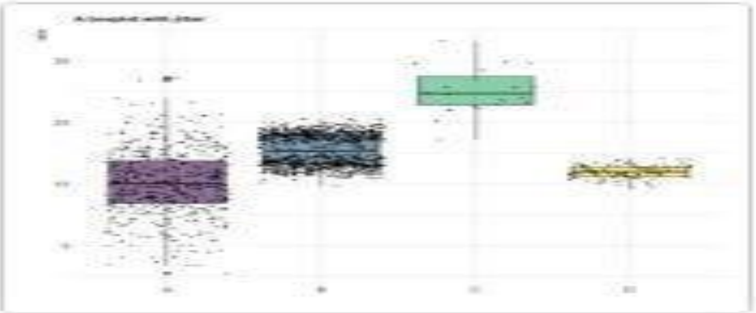


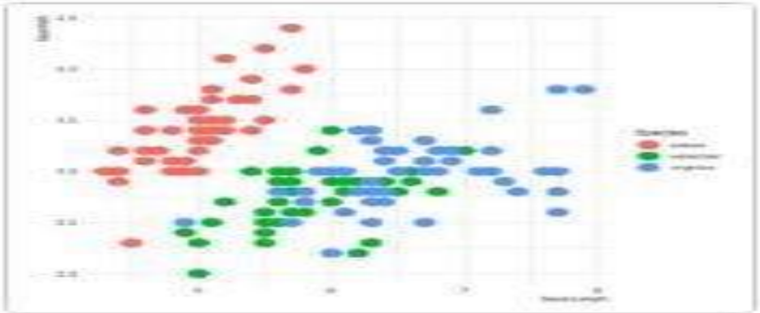

Sepal Length by Species
Comparing distributions across groups
1.2
0.8
0.4
0.0
Density
5
6
7
8
Sepal Length (cm)
Species
setosa
versicolor
virginica

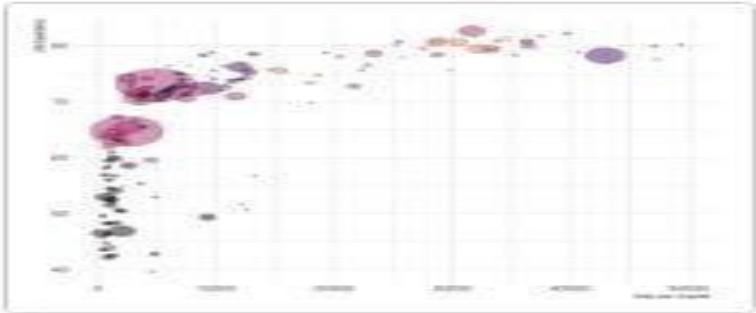


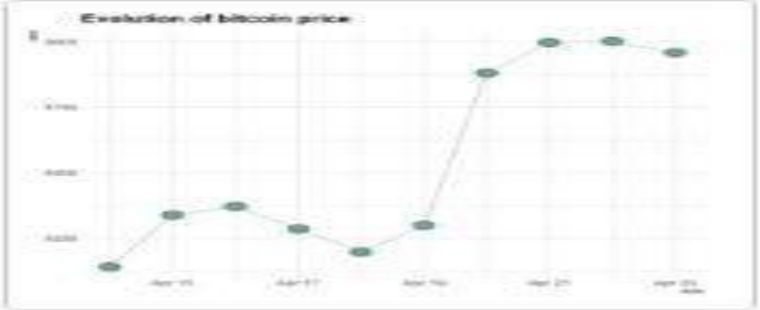
4.5
4.0
3.5
3.0
2.5
2.0
Sepal.Width
5
6
7
8
Sepal.Length
Species
setosa
versicolor
virginica

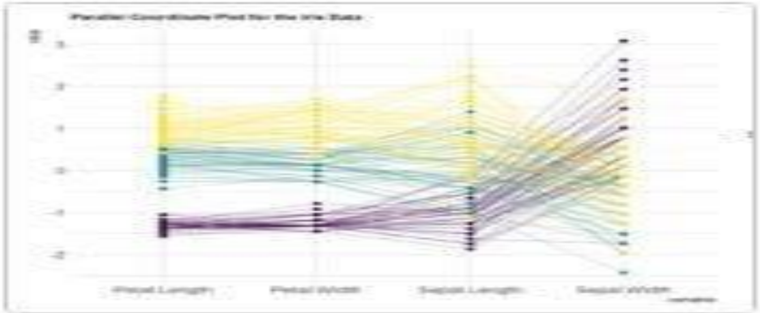

A Violin wrapping a boxplot
30
20
10
0
A
n=500
B
n=1000
C
n=20
D
n=100

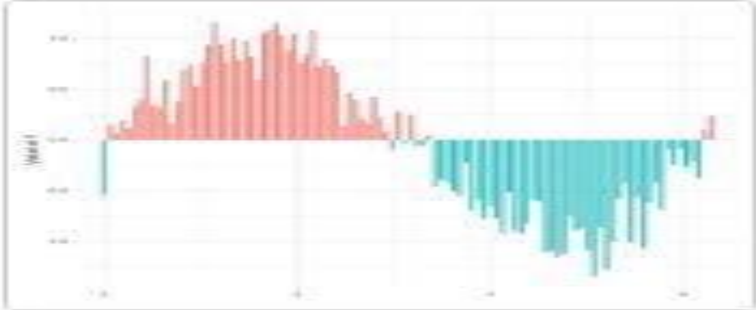


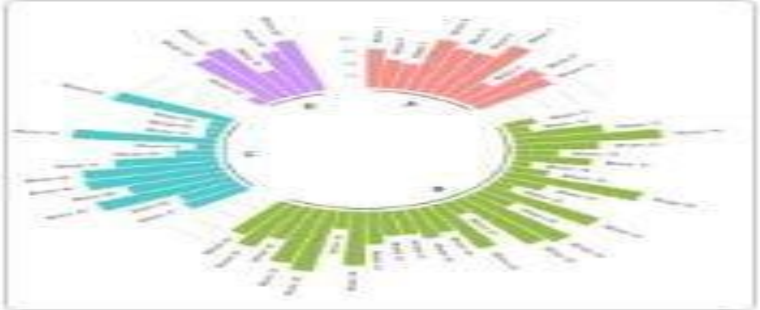


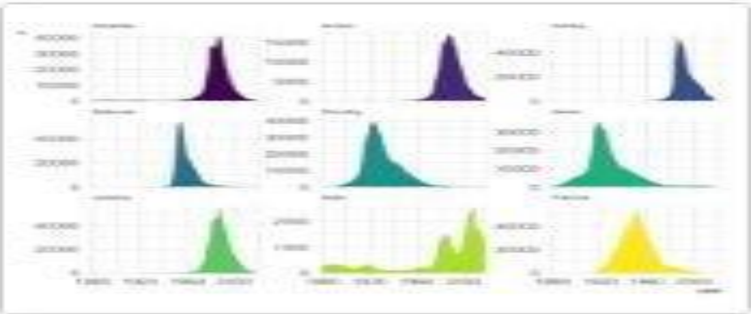


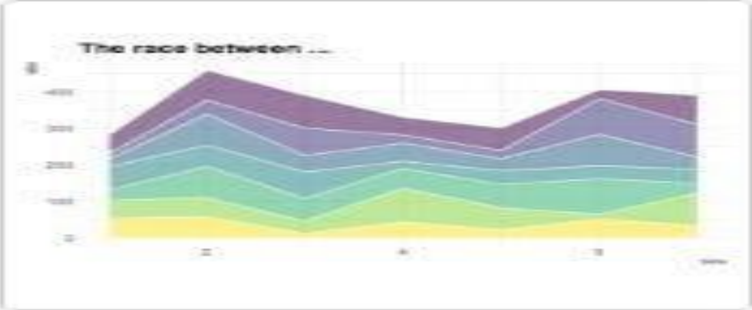

Evolution of bitcoin price


Petal.Length
Petal.Width
Sepal.Length
Sepal.Width








The race between ...

Course Prerequisite

- Basic computer skills
- Programming skill is not required
- Ability to use search engines
- Willingness to learn

What will you Learn?

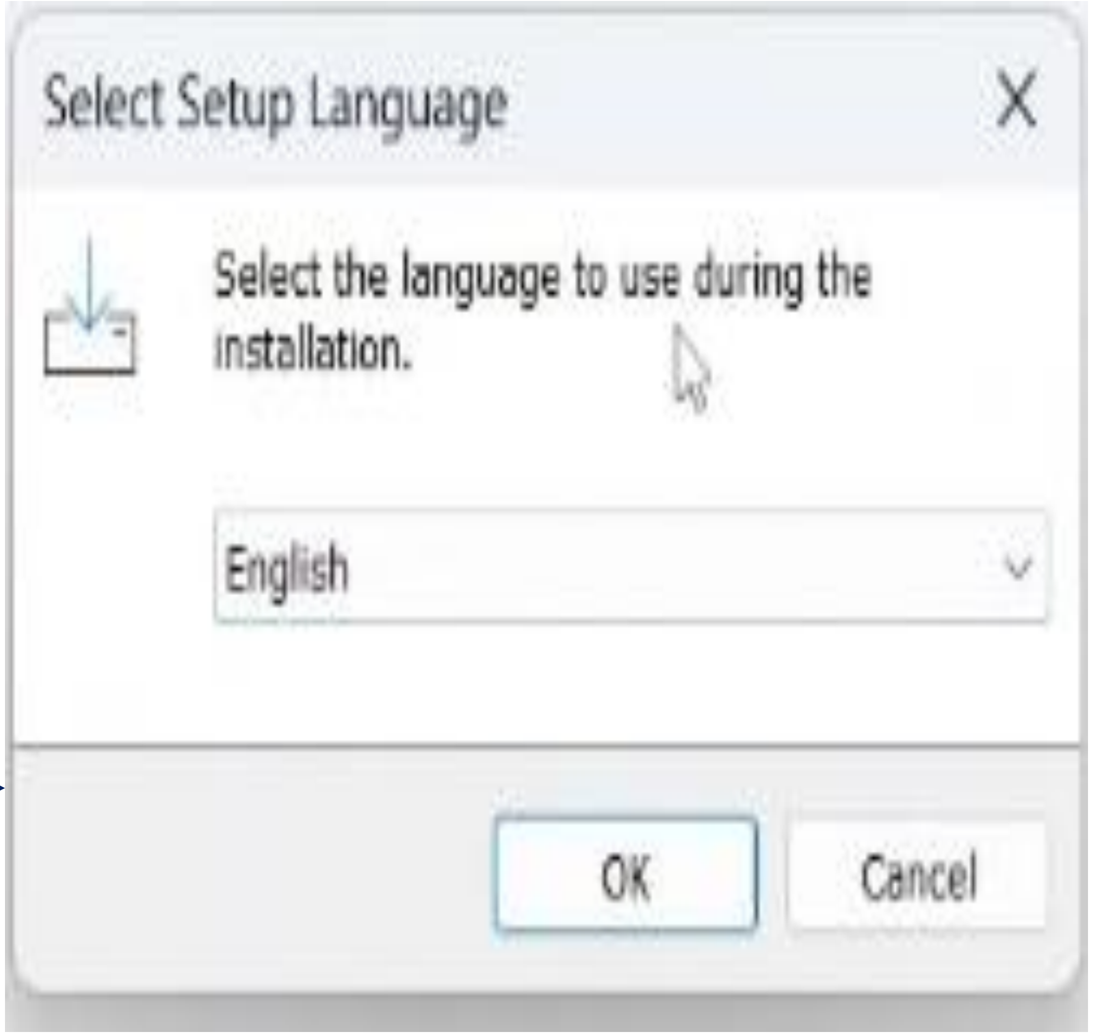
- R Programming Fundamentals
- Data manipulation and visualization
- Applied techniques

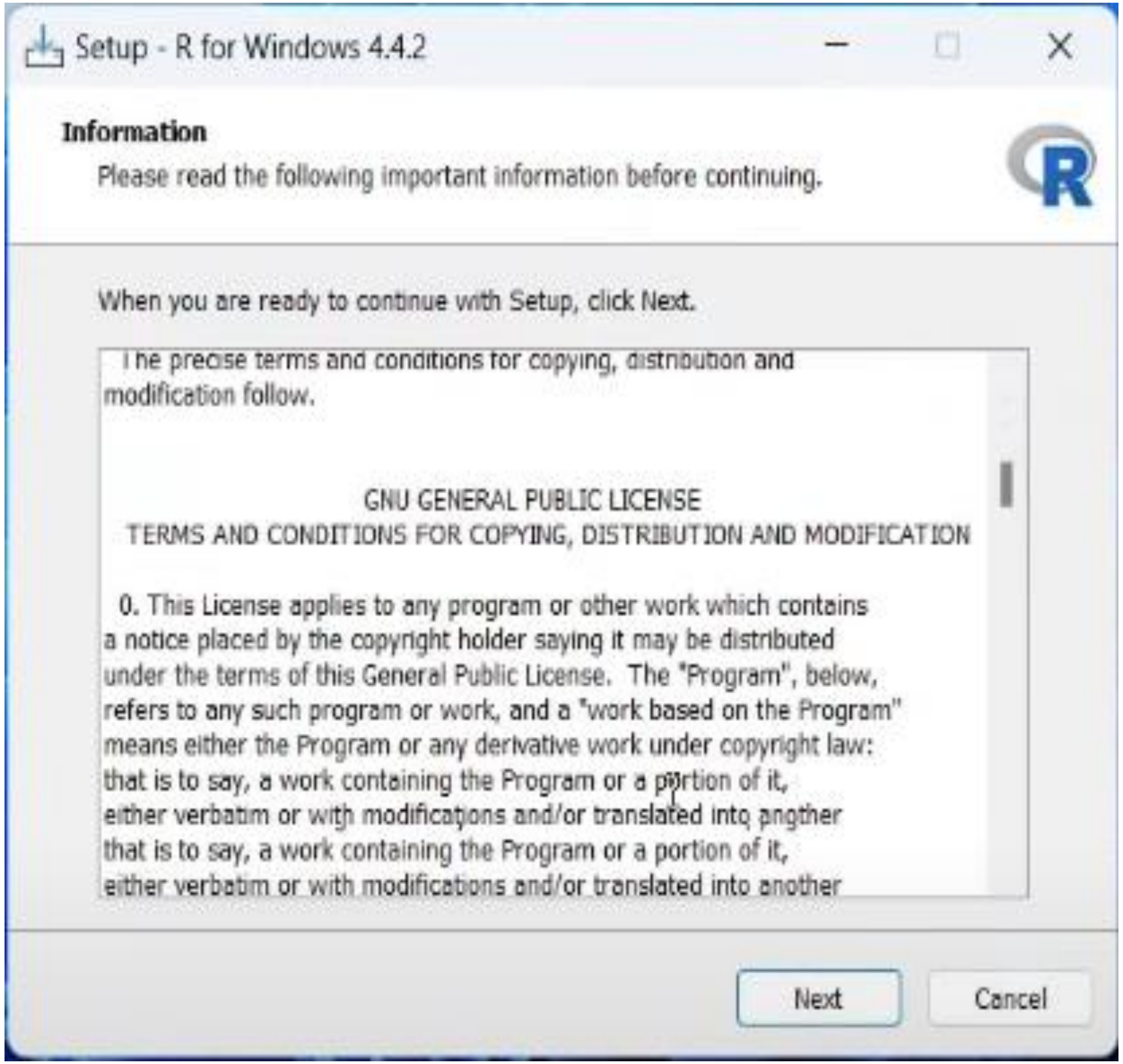
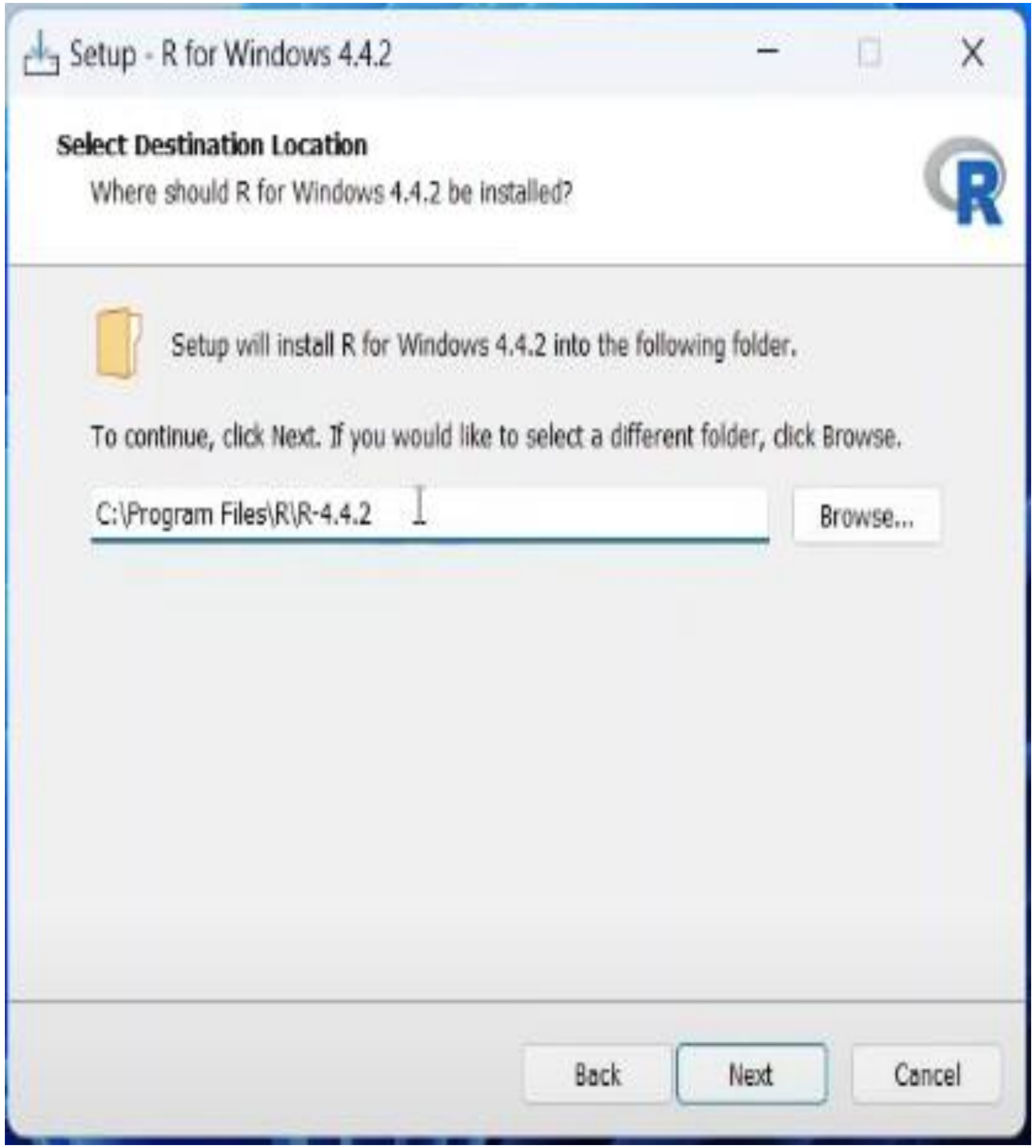
R	RStudio
R is a programming language and software environment.	RStudio is an Integrated Development Environment (IDE) for R.
It is used for statistical computing, data analysis, and visualization.	It provides a user-friendly interface for working with R.
R can run code independently.	RStudio uses R to execute the code.
R is the core engine.	RStudio is the working environment/interface.
R must be installed first.	RStudio is installed after R.

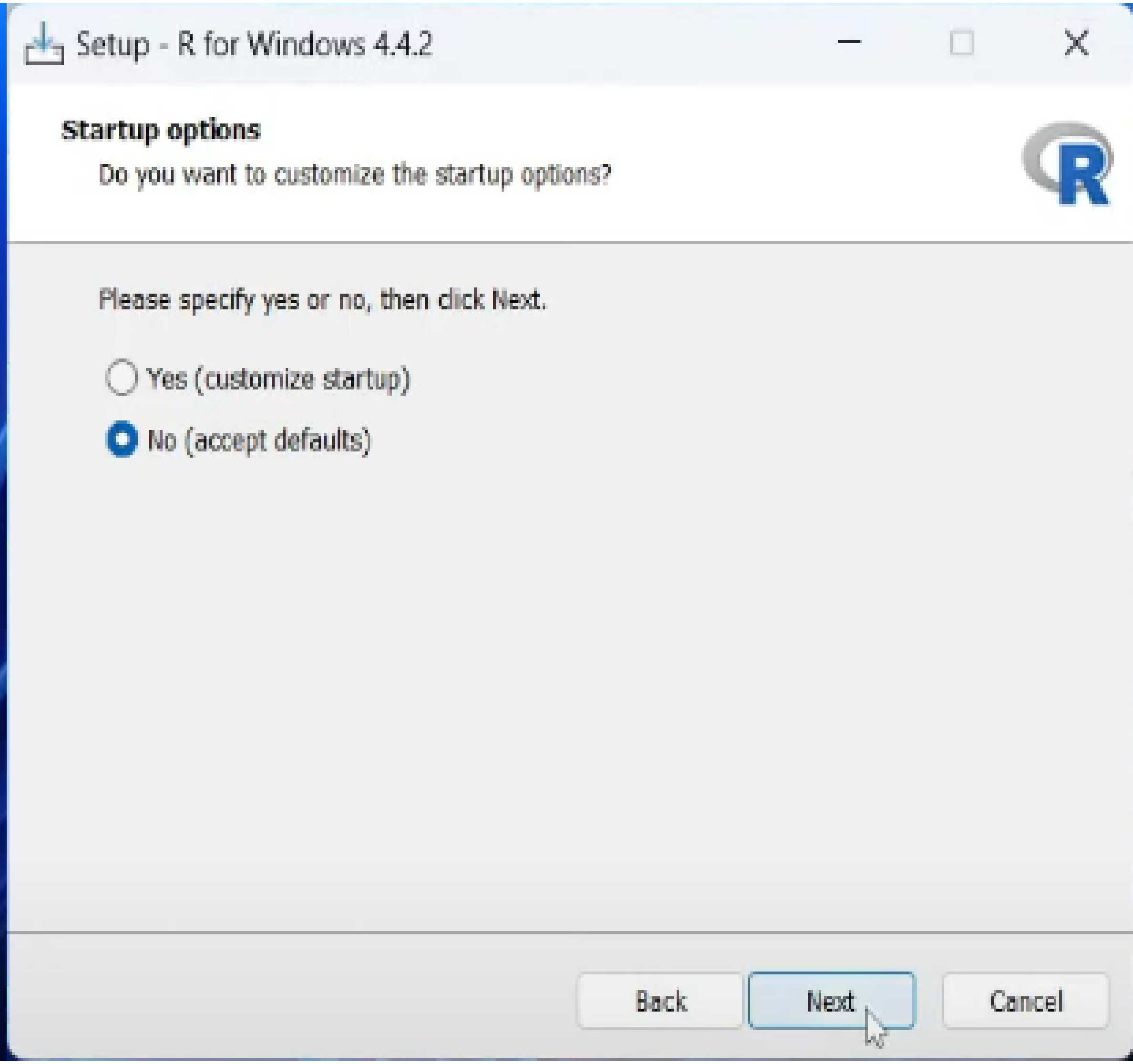
https://cran.rstudio.com/
https://posit.co/download/rstudio-desktop/
https://cran.r-project.org/bin/windows/Rtools/

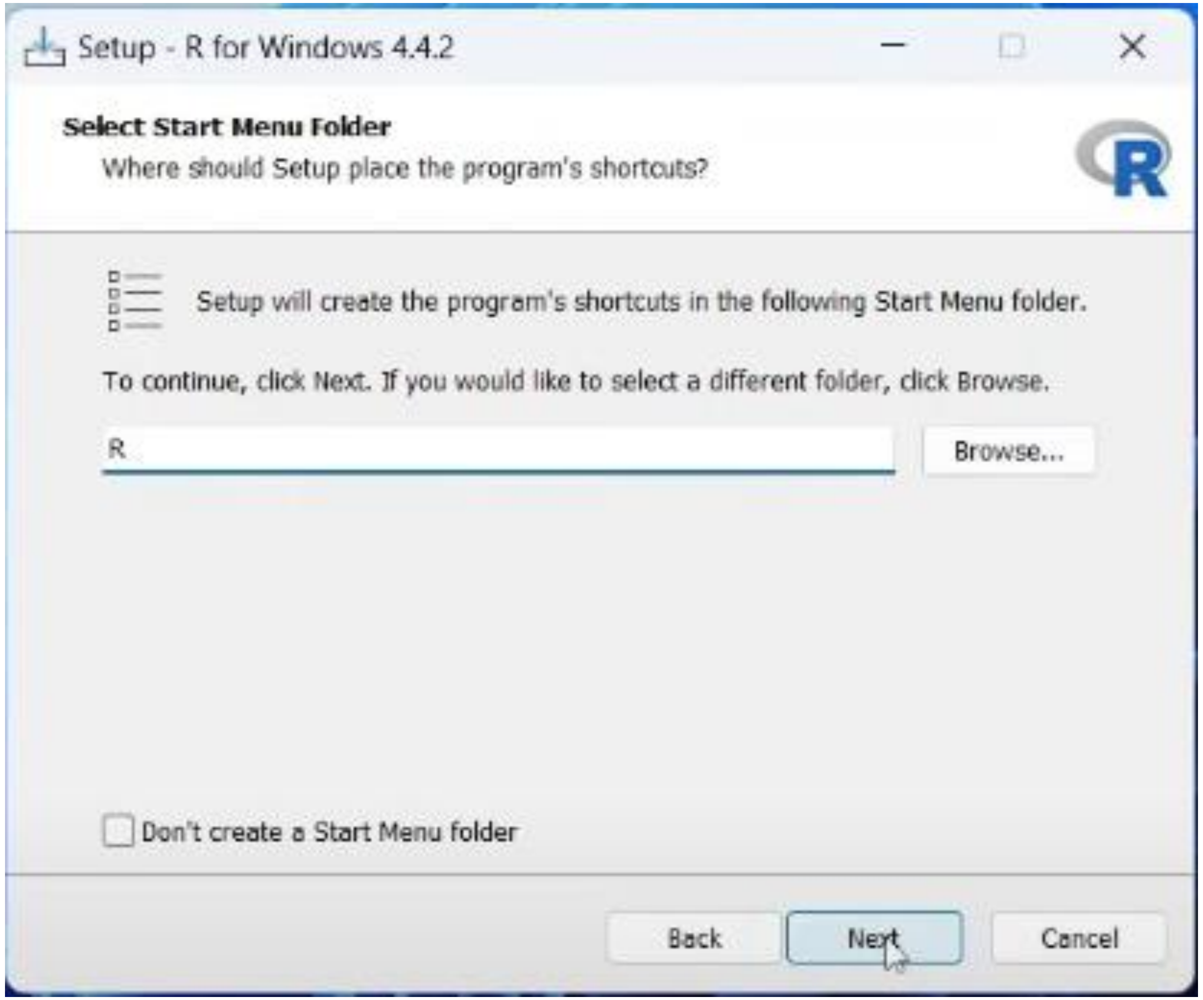
https://posit.co/download/rstudio-desktop/

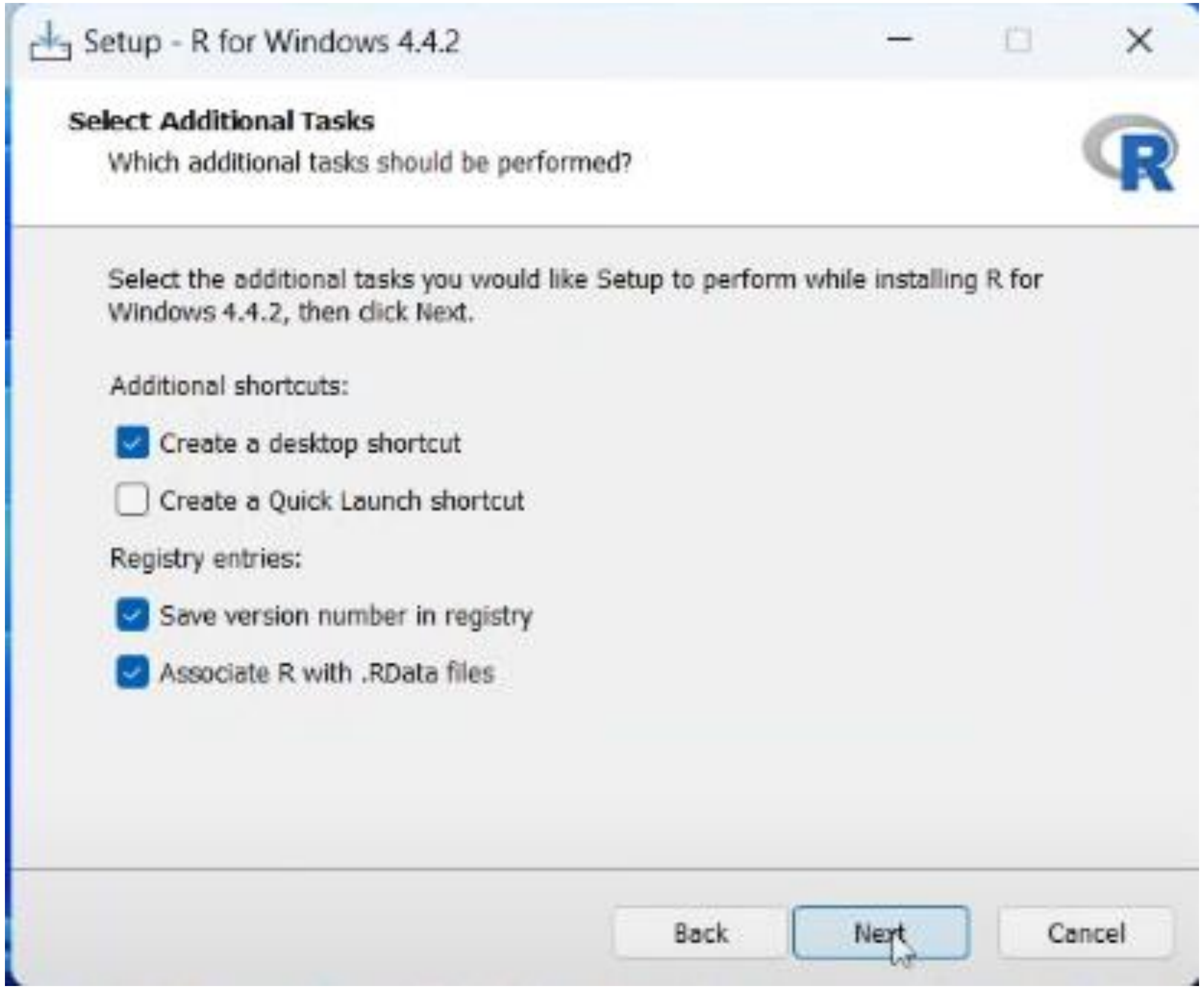
Click on yes

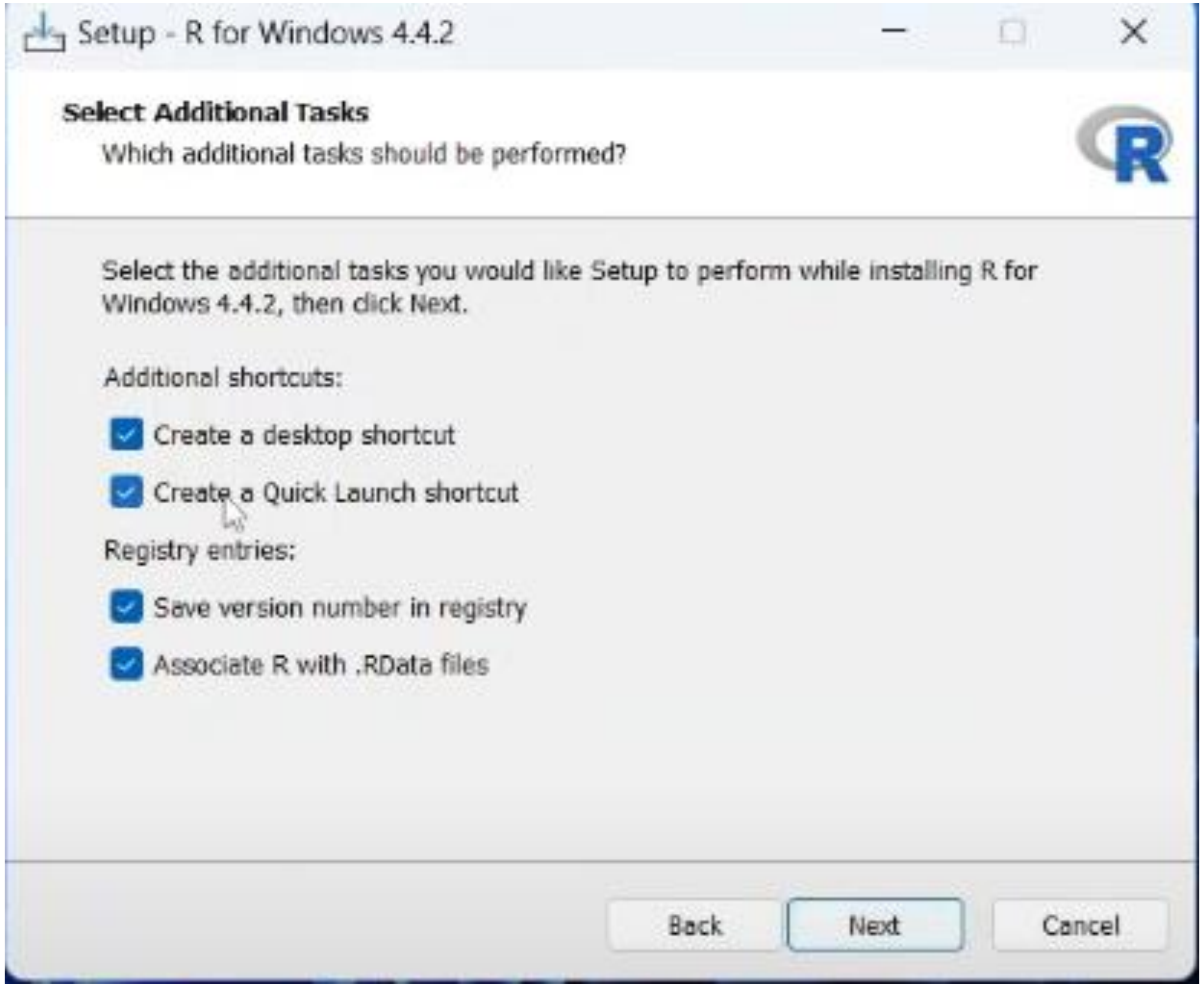



Setup - R for Windows 4.4.2
Information
Please read the following important information before continuing.
When you are ready to continue with Setup, click Next.
The precise terms and conditions for copying, distribution and modification follow.
GNU GENERAL PUBLIC LICENSE
TERMS AND CONDITIONS FOR COPYING, DISTRIBUTION AND MODIFICATION
0. This License applies to any program or other work which contains a notice placed by the copyright holder saying it may be distributed under the terms of this General Public License. The "Program", below, refers to any such program or work, and a "work based on the Program" means either the Program or any derivative work under copyright law: that is to say, a work containing the Program or a portion of it, either verbatim or with modifications and/or translated into another that is to say, a work containing the Program or a portion of it, either verbatim or with modifications and/or translated into another
Next
Cancel

Setup - R for Windows 4.4.2
Select Destination Location
Where should R for Windows 4.4.2 be installed?
Setup will install R for Windows 4.4.2 into the following folder.
To continue, click Next. If you would like to select a different folder, click Browse.
C:\Program Files\R\R-4.4.2
Browse...
Back
Next
Cancel


Setup - R for Windows 4.4.2
Select Components
Which components should be installed?
Select the components you want to install; clear the components you do not want to install. Click Next when you are ready to continue.
User installation
Main Files 92.7 MB
64-bit Files 74.5 MB
Message translations 10.2 MB
Current selection requires at least 180.3 MB of disk space.
Back
Next
Cancel

Setup - R for Windows 4.4.2
Startup options
Do you want to customize the startup options?
Please specify yes or no, then click Next.
Yes (customize startup)
No (accept defaults)
Back
Next
Cancel

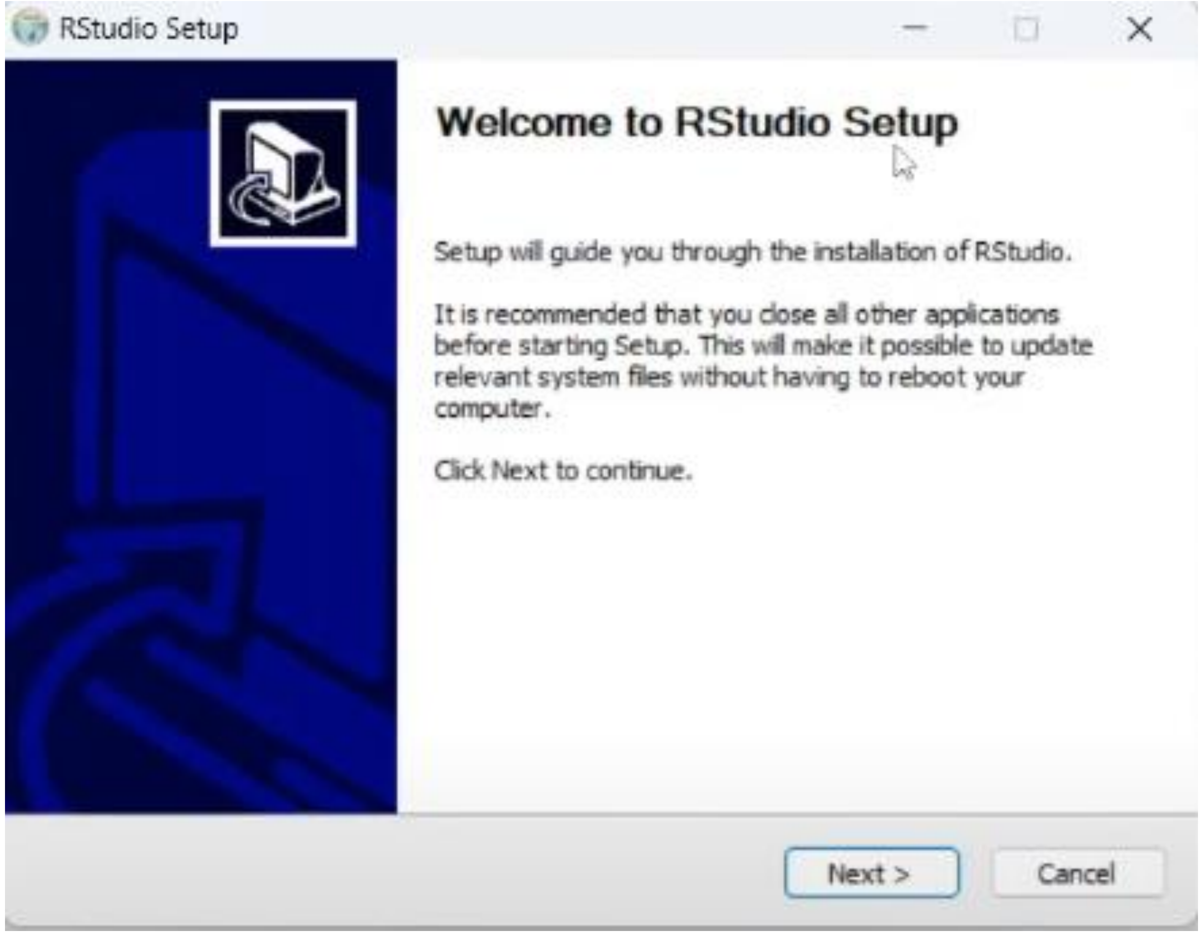

Setup - R for Windows 4.4.2
Select Start Menu Folder
Where should Setup place the program's shortcuts?
Setup will create the program's shortcuts in the following Start Menu folder.
To continue, click Next. If you would like to select a different folder, click Browse.
R
Browse...
Don't create a Start Menu folder
Back
Next
Cancel


Setup - R for Windows 4.4.2
Select Additional Tasks
Which additional tasks should be performed?
Select the additional tasks you would like Setup to perform while installing R for Windows 4.4.2, then click Next.
Additional shortcuts:
Create a desktop shortcut
Create a Quick Launch shortcut
Registry entries:
Save version number in registry
Associate R with .RData files
Back
Next
Cancel

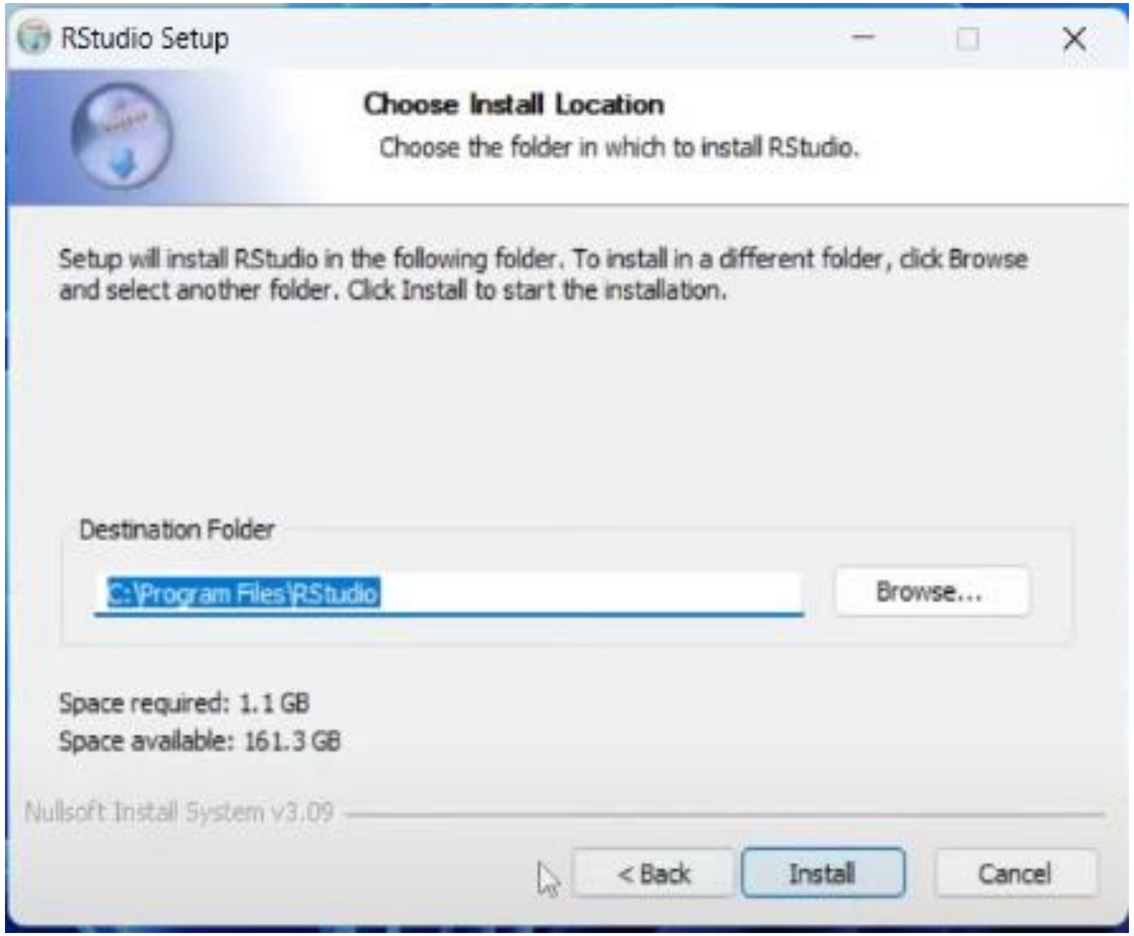


Please wait for a while.

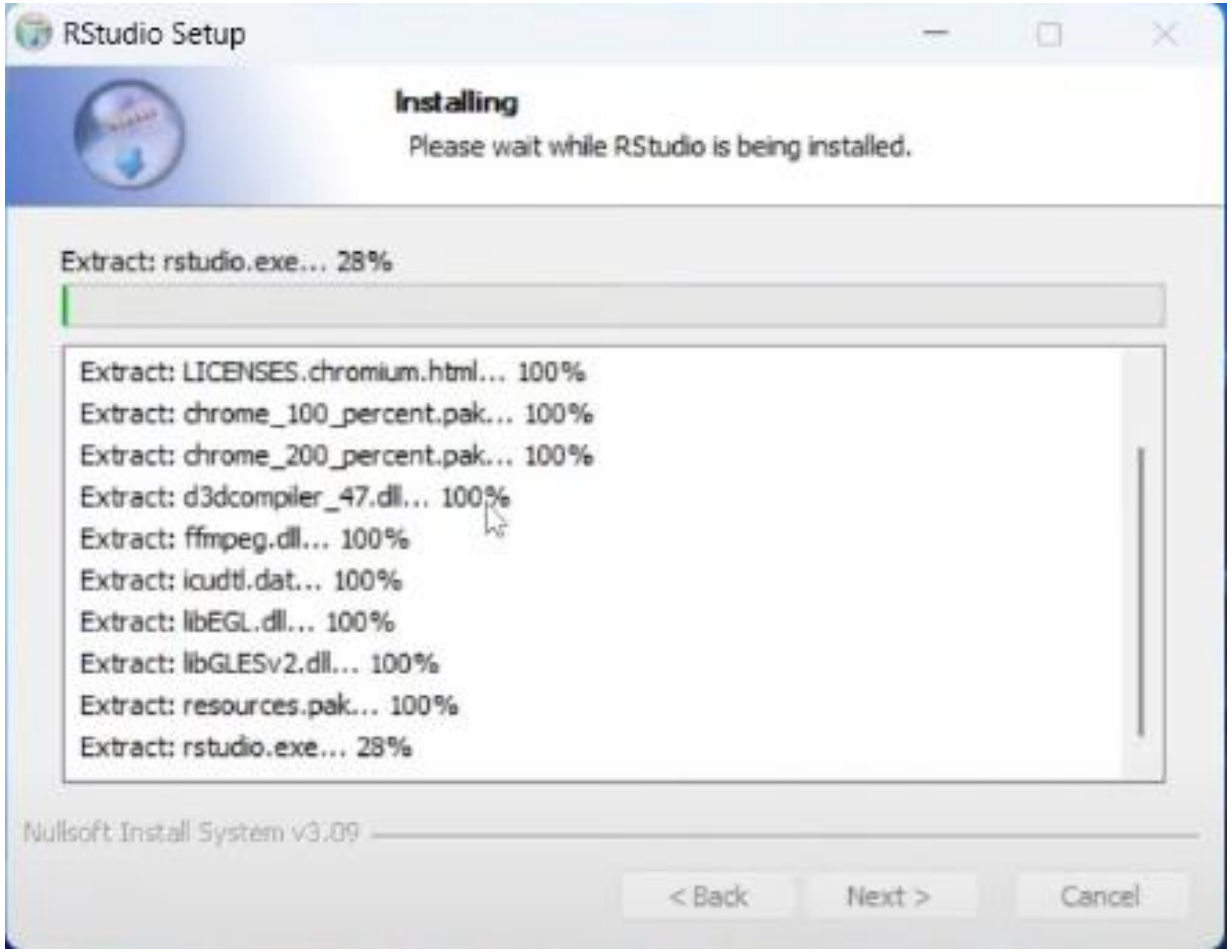

Setup - R for Windows 4.4.2
Completing the R for Windows 4.4.2 Setup Wizard
Setup has finished installing R for Windows 4.4.2 on your computer. The application may be launched by selecting the installed shortcuts.
Click Finish to exit Setup.
Finish

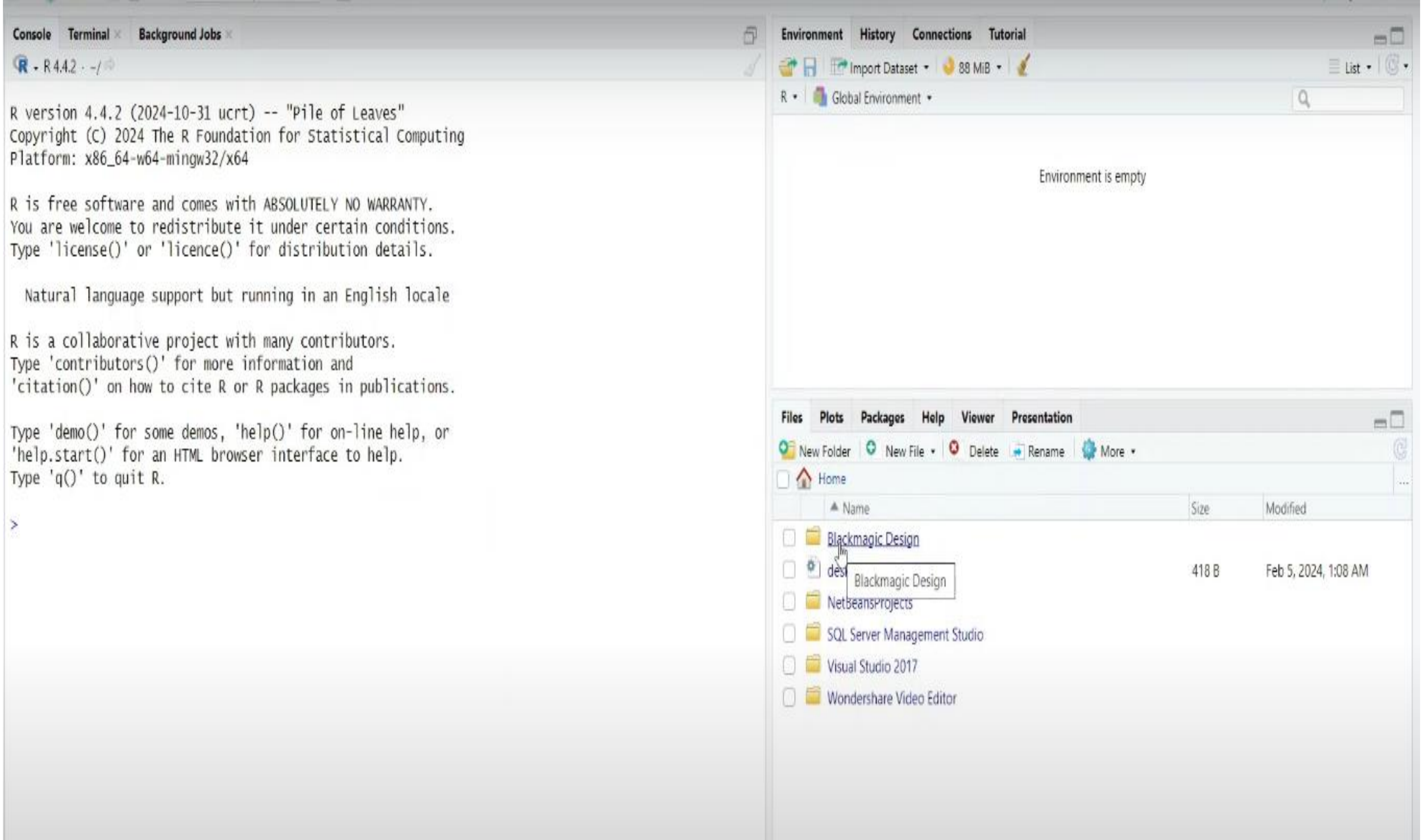


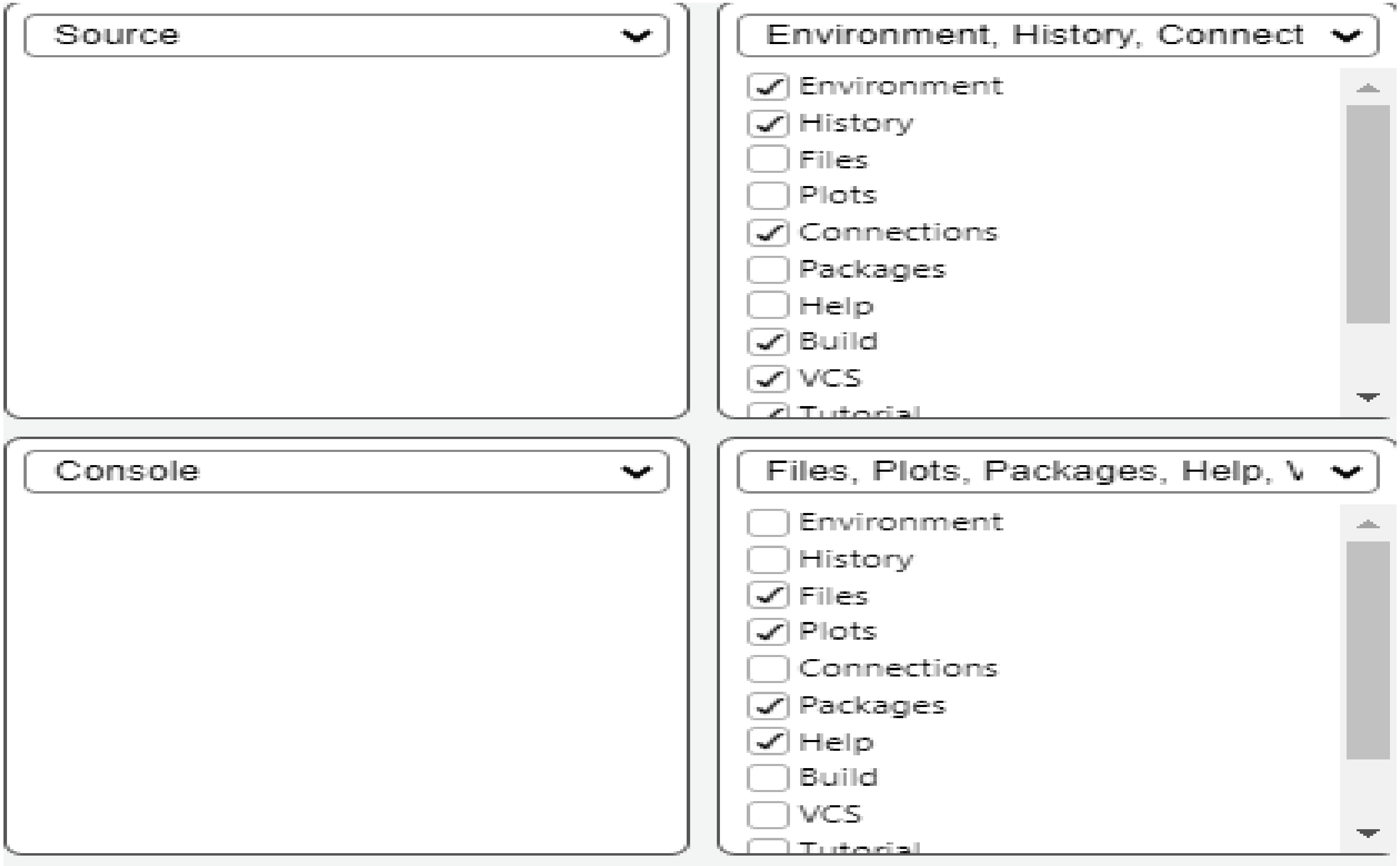
RStudio Setup
Welcome to RStudio Setup
Setup will guide you through the installation of RStudio.
It is recommended that you close all other applications before starting Setup. This will make it possible to update relevant system files without having to reboot your computer.
Click Next to continue.
Next >
Cancel



RStudio Setup
Choose Install Location
Choose the folder in which to install RStudio.
Setup will install RStudio in the following folder. To install in a different folder, click Browse and select another folder. Click Install to start the installation.
Destination Folder
C:\Program Files\RStudio
Browse...
Space required: 1.1 GB
Space available: 161.3 GB
Nullsoft Install System v3.09
< Back
Install
Cancel


RStudio Setup
Installing
Please wait while RStudio is being installed.
Extract: rstudio.exe... 28%
Extract: LICENSES.chromium.html... 100%
Extract: chrome_100_percent.pak... 100%
Extract: chrome_200_percent.pak... 100%
Extract: d3dcompiler_47.dll... 100%
Extract: ffmpeg.dll... 100%
Extract: icudtl.dat... 100%
Extract: libEGL.dll... 100%
Extract: libGLESv2.dll... 100%
Extract: resources.pak... 100%
Extract: rstudio.exe... 28%
Nullsoft Install System v3.09
< Back
Next >
Cancel


Console
Terminal
Background Jobs
R 4.4.2 · ~/
R version 4.4.2 (2024-10-31 ucrt) -- "Pile of Leaves"
Copyright (C) 2024 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64
R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.
Natural language support but running in an English locale
R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.
>
Environment
History
Connections
Tutorial
Import Dataset
88 MiB
List
R
Global Environment
Environment is empty
Files
Plots
Packages
Help
Viewer
Presentation
New Folder
New File
Delete
Rename
More
Home
Name
Size
Modified
Blackmagic Design
418 B
Feb 5, 2024, 1:08 AM
Blackmagic Design
NetBeansProjects
SQL Server Management Studio
Visual Studio 2017
Wondershare Video Editor


Source
Environment, History, Connect
Environment
History
Files
Plots
Connections
Packages
Help
Build
VCS
Console
Files, Plots, Packages, Help,
Environment
History
Files
Plots
Connections
Packages
Help
Build
VCS

2nd Class

- Data types and variables
- Common operators
- Common functions

Common Data Types in R

Data type	Example	typeof()	mode()
Numeric	6, 3.14, 10	double	numeric
Integer	6L, 3L, 10L	integer	numeric
Logical	TRUE, FALSE	logical	logical
Character	“Female” , “Success”, “a”	character	character
Complex	4+3i, 2+9i	complex	complex

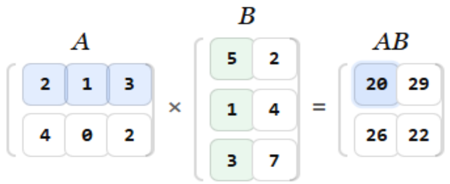
Combine Multiple Elements

- When working with multiple elements of a data type, use the function `c()` to combine multiple elements.

Assignment operators in R

Operators	Example
Leftwards Arrow <-	`x <- 10` `y <- c(3, 2)`
->	`10 -> x`
=	`X = 10`

Arithmetic operators in R

Operators	Description	Example
+	Addition / Sum	10+3 = 13
-	Subtraction / Minus	10-3 = 7
/	Division	10/3 = 3.333333
*	Multiplication	10*3 = 30
^	Exponent / Power	10^3 = 1000
%%	Modulus (Remainder from division)	10%%3 = 1
%/%	Integer division	10%/%3 = 3
%*%	Matrix Multiplication	

Relational operators in R

Operators	Description	Example
<	Less than	4 < 2 FALSE 4 < 10 TRUE
<=	Less than or equal to	4 <=5 FALSE 4 <= 4 TRUE
>	Greater than	4 > 2 TRUE
>=	Greater than or equal to	4 >= 2 FALSE
==	Equal	3 == 5 FALSE
!=	Not equal	3 != 5 TRUE
%in%	Included in	‘a’ %in% c(‘b’, ‘a’, ‘c’) TRUE

Logical operators in R

Operators	Description	Example

&	And	`> x <- c(TRUE, FALSE, TRUE)`
		`> y <- c(TRUE, TRUE, FALSE)`
		`> x & y`
		`[1]  TRUE FALSE FALSE`

|	Or	`> x <- c(TRUE, FALSE, TRUE)`
		`> y <- c(FALSE, TRUE, FALSE)`
		`> x | y`
		`[1] TRUE TRUE TRUE`

!	Not	`> x <- c(TRUE, FALSE, TRUE)`
		`> !x`
		`[1] FALSE  TRUE FALSE`

Note: Relational operators compare values and return either TRUE or FALSE. Logical operators perform logical operations on TRUE and FALSE.

Common functions in R

Group	Functions
Statistical	`sum(), mean(), min(), max(), median(), quantile(), sd(), range(), summary()`
Logical	`is.na(), any(), all(), which()`
Rounding	`round(), floor(), ceiling()`
Other mathematical	`sqrt(), abs(), log(), exp()`
Vector operation	`length(), seq(), rep(), unique()`
Object inspection	`str(), typeof(), mode()`
Utility	`help(), install.packages(), library()`

Thank You